Report on 'TI Surface State under Magnetic Field'

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Abstract

This is a report following letter[1].

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I. 2DEG under Magnetic Field — Landau Gauge

For free charged particle in magnetic field, the Hamiltonian of system is

$$H = \frac{\mathbf{\Pi}^2}{2m} = \frac{(\boldsymbol{p} - q\boldsymbol{A})^2}{2m},\ [x_i, p_j] = i\hbar. \tag{1}$$

Now say that, for free electron with $q = -e$ ($e > 0$) moving in 2-D $x - y$ plane with vertical uniform magnetic field $\boldsymbol{B} = B\hat{z}$ ($B > 0$), one has corresponding vector potential $\boldsymbol{A}$ as $\nabla \times \boldsymbol{A} = \boldsymbol{B}$, then immediately,

$$[\Pi_x, \Pi_y] = [p_x + eA_x, p_y + eA_y] = -ie\hbar(\nabla \times \boldsymbol{A})_z = -ie\hbar B, \tag{2}$$

thus

$$[\Pi_x - i\Pi_y, \Pi_x + i\Pi_y] = 2i[\Pi_x, \Pi_y] = 2e\hbar B, \tag{3}$$

from which one could define Bosonic annihilation and creation operators as

$$a = \frac{1}{\sqrt{2e\hbar B}}(\Pi_x - i\Pi_y),\ a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(\Pi_x + i\Pi_y),\ [a, a^\dagger] = 1. \tag{4}$$

Then one has

$$H = \hbar\omega(a^\dagger a + \frac{1}{2}),\ \omega = \frac{eB}{m}, \tag{5}$$

and clearly it will own the same algebra structure like a harmonic oscillator with

$$N = a^\dagger a,\ N\left|n\right\rangle = n\left|n\right\rangle,\ H\left|n\right\rangle = \hbar\omega(n + \frac{1}{2})\left|n\right\rangle, \tag{6}$$

w with each $|n\rangle$ forming energy level known as the *Landau level*. The states are bounded and constructed by

$$a\left|0\right\rangle = 0,\ \left|n\right\rangle = \frac{1}{\sqrt{n!}}(a^\dagger)^n\left|0\right\rangle, \tag{7}$$

with

$$a\left|n\right\rangle = \sqrt{n}\left|n-1\right\rangle,\ a^\dagger\left|n\right\rangle = \sqrt{n+1}\left|n+1\right\rangle. \tag{8}$$

The above relations hold for any gauge, and now we choose Landau gauge $\boldsymbol{A} = (-By, 0, 0)$, which means that

$$\begin{cases} \Pi_x = p_x - eBy,\ \Pi_y = p_y, \end{cases} \tag{9}$$

which preserves commutative relation between p_x and Hamiltonian thus yields $p_x = \hbar k_x$ with k_x a good quantum number, and Landau levels are now composited states $|n, k_x\rangle$. What is more, the annihilation and creation operators under the position representation now read

$$a = \frac{1}{\sqrt{2e\hbar B}}(p_x - eBy - ip_y) = \frac{1}{\sqrt{2}}[(y_0 - y)/l_B - l_B\partial_y], \tag{10a}$$

$$a^\dagger = \frac{1}{\sqrt{2e\hbar B}}(p_x - eBy + ip_y) = \frac{1}{\sqrt{2}}[(y_0 - y)/l_B + l_B\partial_y], \tag{10b}$$

where

$$l_B^2 = \frac{\hbar}{m\omega} = \frac{\hbar}{eB},\ y_0 = k_x l_B^2.$$

Now in this case, one may write

$$H = \frac{(eB)^2}{2m}\left[(y - l_B^2 k_x)^2 + \frac{p_y^2}{(eB)^2}\right] = \frac{p_y^2}{2m} + \frac{1}{2}m\omega^2(y - l_B^2 k_x)^2, \tag{11}$$

which is a harmonic oscillator with central shifted by $y_0 = l_B^2 k_x$. For its wavefunction, firstly notice the ground state bound by $\langle r|a|0, k_x\rangle = 0$, which leads to the equation

$$\left[\partial_y + \frac{y - y_0}{l_B^2}\right]\varphi_{0,k_x}(r) = 0, \tag{12}$$

which offers us the normalized solution as

$$\psi_{0,k_x}(r) = \frac{1}{\sqrt{L_x}}\mathrm{e}^{ik_x x}\frac{1}{\pi^{1/4}\sqrt{l_B}}\exp\left(-\frac{1}{2}\left(\frac{y - y_0}{l_B}\right)^2\right), \tag{13}$$

with L_x the system length along x direction, and the factor $1/\sqrt{L_x}$ provides proper normalization since now

$$\langle k_x|k'_x\rangle = \int_0^{L_x} \mathrm{d}x \, \langle k_x|x\rangle \, \langle x|k'_x\rangle = \frac{1}{L_x}\int_0^{L_x} \mathrm{d}x e^{i(k'_x-k_x)x} = \delta_{k_x,k'_x},$$

while the factor $1/\pi^{1/4}\sqrt{l_B}$ normalizes Guassian like $n=0$ state. For the higher state, the iteration relation tells that its wavefunction reads

$$\begin{aligned}
\varphi_{n,k_x}(r) &= \langle r|n,k_x\rangle = \langle x|k_x\rangle \, \langle y| \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle \\
&= \frac{1}{\sqrt{L_x}}\mathrm{e}^{ik_x x}\frac{1}{\sqrt{n!}}\left(\frac{1}{\sqrt{2}}[(y_0-y)/l_B + l_B\partial_y]\right)^n \frac{1}{\pi^{1/4}\sqrt{l_B}}\exp\left(-\frac{1}{2}\left(\frac{y-y_0}{l_B}\right)^2\right) \\
&= \frac{1}{\sqrt{L_x l_B}}e^{ik_x x}\frac{1}{\sqrt{2^n n!\sqrt{\pi}}}(-z+\partial_z)^n e^{-z^2/2}, \ z \equiv (y-y_0)/l_B, \\
&= \frac{1}{\sqrt{L_x l_B}}e^{ik_x x}\frac{(-1)^n}{\sqrt{2^n n!\sqrt{\pi}}}e^{-z^2/2}(2z-\partial_z)^n \cdot 1 = \frac{1}{\sqrt{L_x l_B}}e^{ik_x x}\phi_n(z),
\end{aligned} \tag{14}$$

where

$$\phi_n(z) = \frac{(-1)^n}{\sqrt{2^n n!\sqrt{\pi}}}e^{-z^2/2}H_n(z), \tag{15}$$

is the *Hermite-Gaussian function* with

$$H_n(z) = (2z-\partial_z)^n \cdot 1 \tag{16}$$

the *Hermite polynomial*. Notice that there exists an additional $(-1)^n$ factor compared with normal harmonic oscillator wavefunction, which is caused by the chosen sign in annihilation and creation operators, and since it would not matter in the particle-number conserved system to do a global phase transition, it is proper to give an $e^{i\pi}$ rotation to annihilation operator which equals add a -1 sign to a and $a^\dagger$, which will cancel the $(-1)^n$ sign, but we preserve that sign here. Also notice that when deriving the last equality, identity

$$f[l(x)+\partial_x]g(x) = g(x)f[l(x)+h(x)+\partial_x]\cdot 1 \tag{17}$$

has been used, with f, l, g arbitrary functions but constrained by $\partial_x g(x) = h(x)g(x)$. This identity could be proved by a direct Taylor expansion

$$\begin{aligned} f[l(x)+\partial_x]g(x) &= \sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!}[l(x)+\partial_x]^n[g(x)\cdot 1] \\ &= \sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!}[l(x)+\partial_x]^{n-1}[l(x)g(x)+\partial_x g(x)+g(x)\partial_x \cdot 1] \\ &= \sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!}[l(x)+\partial_x]^{n-1}\left\{g(x)[l(x)+h(x)+\partial_x]\cdot 1\right\} \\ &= \cdots = \sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!} g(x)[l(x)+h(x)+\partial_x]^n \cdot 1 = g(x) f[l(x)+h(x)+\partial_x]\cdot 1, \end{aligned}$$

and notice that $\cdot 1$ at the end is not necessary but for recognition of the action object with respect to ∂_x. The identity could be generalized to operator form with pre-assumption that $[l, g] = 0 = [h, g]$.

Some properties of Hermite polynomial

Firstly by the identity above, Hermite polynomial has another equivalent expression as

$$H_n(x) = (2x - \partial_x)^n \cdot 1 = (-1)^n e^{x^2} \partial_x^n e^{-x^2}.$$

Then by notice that

$$\begin{aligned} \partial_x^{n+1} e^{-x^2} &= \partial_x^n(-2xe^{-x^2}) = C_n^0(-2x)\partial_x^n e^{-x^2} + C_n^1[\partial_x(-2x)]\partial_x^{n-1} e^{-x^2} \\ &= -2x\partial_x^n e^{-x^2} - 2n\partial_x^{n-1} e^{-x^2}, \end{aligned}$$

one recognizes the recurrent relation counting on $(-1)^n$ factor as

$$H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) = 0.$$

With this relation, it is straightforward to verify that Hermite polynomial satisfies the differential equation

$$H''(x) - 2xH'(x) + 2nH(x) = 0, \ \forall n \in \mathbb{N},$$

which lies at the heart of analytic harmonic oscillator problem. Meanwhile, we have

$$H_n'(x) = 2xH_n(x) - H_{n+1}(x) = 2nH_{n-1}(x).$$

For the desired orthonormalization relation of Hermite polynomial, notice that it is generated by

$$e^{2xt-t^2} = \sum_{n=0}^{+\infty} H_n(x)\frac{t^n}{n!},$$

since

$$\partial_t^n e^{2xt-t^2}\big|_{t=0} = \partial_t^n e^{-(t-x)^2+x^2}\big|_{t=0} = (-1)^n e^{x^2}\partial_x^n e^{-(t-x)^2}\big|_{t=0} = (-1)^n e^{x^2}\partial_x^n e^{-x^2} = H_n(x).$$

Then consider a square form

$$e^{2xt-t^2}e^{2xs-s^2} = \sum_{m,n} H_n(x)H_m(x)\frac{t^n s^m}{n!m!},$$

with weighted measure integral

$$\int_{\mathbb{R}} \mathrm{d}x e^{2xt-t^2}e^{2xs-s^2}e^{-x^2} = e^{2st}\int_{\mathbb{R}} \mathrm{d}x e^{-[x-(s+t)]^2} = \sqrt{\pi}\sum_n \frac{2^n}{n!}s^n t^n,$$

compared with right-hand-side form to conclude

$$\int_{\mathbb{R}} \mathrm{d}x H_n(x)H_m(x)e^{-x^2} = \sqrt{\pi}2^n n!\delta_{mn},$$

as orthonormality, or simply $\langle H_n, H_m\rangle_w = \sqrt{\pi}2^n n!\delta_{mn}$ with weighted inner product.

With orthonormality and recurrent relation above, we could derive inner product between ϕ_n with the usual un-weighted definition

$$\langle f, g\rangle = \int_{\mathbb{R}} \mathrm{d}x f(x)g(x),$$

and note that functions here are all chosen real without need of complex conjugate. Now

$$\begin{aligned}
\langle \phi_m, \phi_n\rangle &= \frac{(-1)^{m+n}}{\sqrt{2^m m!2^n n!\pi}}\langle H_m, H_n\rangle_w = (-1)^{m+n}\delta_{mn} = \delta_{mn},\\
\langle x\phi_m, \phi_n\rangle &= \frac{(-1)^{m+n}}{\sqrt{2^m m!2^n n!\pi}}\langle xH_m, H_n\rangle_w = \frac{(-1)^{m+n}}{\sqrt{2^m m!2^n n!\pi}}\langle \frac{1}{2}H_{m+1} + mH_{m-1}, H_n\rangle\\
&= -\sqrt{\frac{m+1}{2}}\delta_{m+1,n} - \sqrt{\frac{m}{2}}\delta_{m-1,n},\\
\langle \phi_m', \phi_n\rangle &= \langle -x\phi_m, \phi_n\rangle + \frac{(-1)^{m+n}}{\sqrt{2^m m!2^n n!\pi}}\langle H_m', H_n\rangle_w = \frac{(-1)^{m+n}}{\sqrt{2^m m!2^n n!\pi}}\langle -\frac{1}{2}H_{m+1} + mH_{m-1}, H_n\rangle_w\\
&= \sqrt{\frac{m+1}{2}}\delta_{m+1,n} - \sqrt{\frac{m}{2}}\delta_{m-1,n},
\end{aligned}$$

which serve as basic neighbour inner-product relation useful in problems about Dirac particle under magnetic field we touch later.

II. Surface State of STI under Magnetic Field — Landau Gauge

2D surface state of an 3D STI holding single Dirac fermion could be written as[2]

$$H(\boldsymbol{k}) = \lambda[\sin(k_x a)\sigma_x + \sin(k_y a)\sigma_y] + \Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k})\sigma_z, \tag{18}$$

where $m_0(\boldsymbol{k}) = m_0 - 4t_{\parallel}\left(\sin^2\frac{k_x a}{2} + \sin^2\frac{k_y a}{2}\right)$. The area $m_0(\boldsymbol{k}) > 0$ defines the parity-invariant-regime[3], and when Fermi surface is endowed inside, the Hall conductance for this regulated single Dirac fermion is quantized to 0.5 quantum unit conductance, thus belongs to a new type of topological class as quantum anomalous metal phase[4]. Such half-quantization nature is mathematically related to the Atiyah-Patodi-Singer η index theorem[5, 6].

To contact the problem of behaviour of such a single Dirac fermion under finite magnetic field, we firstly translate such a model into its continuum version as

$$H(\boldsymbol{p}) = v\boldsymbol{p}\cdot\sigma + M(p^2)\sigma_z, \tag{19}$$

where $M(p^2) = \Theta(-m_0(p))m_0(p)$, and $m_0(p) = m_0 - bp^2 = b(p_c^2 - p^2) = b\hbar^2(k_c^2 - k^2)$, $\boldsymbol{p} = \hbar\boldsymbol{k}$ with new definition

$$\begin{cases} v\hbar = \lambda a \\ b\hbar^2 = t_{\perp}a^2 \\ m_0 = b\hbar k_c^2 \end{cases}. \tag{20}$$

And by minimal coupling with a static magnetic field $\boldsymbol{p} \to \boldsymbol{\Pi} = \boldsymbol{p} + e\boldsymbol{A}$ with $e > 0$ the unit charge, we have

$$H(\boldsymbol{\Pi}) = \begin{pmatrix} M(\Pi^2) & v(\Pi_x - i\Pi_y) \\ v(\Pi_x + i\Pi_y) & -M(\Pi^2) \end{pmatrix}. \tag{21}$$

Since definition of physical momentum here is the same as with free charged particle, we also recognize by setting perpendicular up magnetic field $\boldsymbol{B} = B\hat{z},\ B > 0$

$$[\Pi_x, \Pi_y] = -ieB\hbar \Rightarrow [a, a^{\dagger}] = 1, \tag{22}$$

with

$$a = \frac{l_B}{\hbar}\frac{1}{\sqrt{2}}(\Pi_x - i\Pi_y),\ a^{\dagger} = \frac{l_B}{\hbar}\frac{1}{\sqrt{2}}(\Pi_x + i\Pi_y), \tag{23}$$

where magnetic length is defined as

$$l_B \equiv \sqrt{\frac{\hbar}{eB}} \approx \frac{25.6\text{nm}}{\sqrt{B[T]}}. \tag{24}$$

And by number operator $N = a^\dagger a$,

$$N\left|n\right\rangle = n\left|n\right\rangle,\ a\left|n\right\rangle = \sqrt{n}\left|n-1\right\rangle,\ a^\dagger\left|n\right\rangle = \sqrt{n+1}\left|n+1\right\rangle, \tag{25}$$

bounded and constructed by

$$a\left|0\right\rangle = 0,\ \left|n\right\rangle = \frac{(a^\dagger)^n}{n!}\left|0\right\rangle. \tag{26}$$

Since now

$$\Pi^2 = \frac{\hbar^2}{l_B^2}(2a^\dagger a + 1), \tag{27}$$

the Hamiltonian becomes

$$H(a, a^\dagger) = \begin{pmatrix} M\left[\frac{\hbar^2}{l_B^2}(2a^\dagger a + 1)\right] & \eta a \\ \eta a^\dagger & -M\left[\frac{\hbar^2}{l_B^2}(2a^\dagger a + 1)\right] \end{pmatrix}, \tag{28}$$

with cyclotron energy defined as

$$\eta = \frac{\sqrt{2}v\hbar}{l_B}. \tag{29}$$

For the eigenvalue problem $H(a, a^\dagger)\left|\Psi\right\rangle = E\left|\Psi\right\rangle$, it is fairly-well to guess the solution considering the off-diagonal single annihilation/creation operator as

$$\left|\Psi_n\right\rangle = \begin{pmatrix} c_{n1}\left|n-1\right\rangle \\ c_{n2}\left|n\right\rangle \end{pmatrix}, \tag{30}$$

which gives equation about the coefficients bringing back to eigen-equation

$$\begin{pmatrix} M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] - E_n & \eta\sqrt{n} \\ \eta\sqrt{n} & -M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right] - E_n \end{pmatrix}\begin{pmatrix} c_{n1} \\ c_{n2} \end{pmatrix} = 0, \tag{31}$$

and gives eigenvalue equation considering non-trivial solution as

$$\left\{M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] - E_n\right\}\left\{-M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right] - E_n\right\} - \eta^2 n = 0. \tag{32}$$

Solving the equation above leads to

$$\varepsilon_{ns} = \frac{\omega_n}{2} + s\sqrt{m_n^2 + \eta^2 n},\ n \geq 1,\ s = \pm, \tag{33}$$

where

$$\omega_n \equiv M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] - M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right], \tag{34a}$$

$$m_n \equiv \frac{1}{2}\left\{ M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] + M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right]\right\}. \tag{34b}$$

For $n=0$, we can define $|-1\rangle = 0$ and $M\left[\frac{\hbar^2}{l_B^2}(-1)\right] = 0$ so that $(0, |0\rangle)^T$ is directly the eigenstate for $n=0$, whose energy is given by $-M\left[\frac{\hbar^2}{l_B^2}(1)\right] = \frac{\omega_0}{2} - \sqrt{m_0^2}$, i.e., the eigenvalue expression also holds for $n=0$ state but with $s=-1$ only, and one needs $|c_{0,2}| = 1$ for normalization.

Notice that by defining $\omega = 2b\hbar^2/l_B^2$, $m_0 = b\hbar^2 k_c^2 = n_c\omega$ with $n_c = m_0/\omega = (k_c l_B)^2/2$, we see that

$$M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right] = \Theta\left[\omega(n+1/2-n_c)\right]\omega(n_c-n-1/2),$$

then depend on three situations, we obtain

• $n > n_c + 1/2$,

$$\omega_n = \omega[(n_c - n + 1/2) - (n_c - n - 1/2)] = \omega,$$
$$m_n = \frac{1}{2}\omega[(n_c - n + 1/2) + (n_c - n - 1/2)] = \omega(n_c - n);$$

• $n < n_c - 1/2$,

$$\omega_n = 0 = m_n;$$

• $n_c - 1/2 < n < n_c + 1/2$, this condition always exists once, denoted as n_t,

$$\omega_{n_t} = -2m_{n_t} = -\omega(n_c - n_t - 1/2) \equiv 2\delta n\omega,\ \ \delta n \equiv (n_t + 1/2 - n_c)/2 > 0;$$

and those conditions lead to

$$\varepsilon_{ns} = \begin{cases} s\sqrt{n}\eta, & n < n_t \\ \omega\delta n + s\sqrt{\omega^2\delta n^2 + \eta^2 n}, & n = n_t \\ \frac{\omega}{2} + s\sqrt{(n-n_c)^2\omega^2 + \eta^2 n}, & n > n_t \end{cases}. \tag{35}$$

Based on discussion above, we now calculate the Hall conductance of the single Dirac fermion in the clean limit utilizing the Kubo-Streda-Bastin formula[7]

$$\sigma_{\alpha\beta} = \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\left[\mathrm{Tr}\big(v_\alpha\partial_\varepsilon G^R v_\beta\,\mathrm{Im}\,G^R\big) - \mathrm{Tr}\big(v_\alpha\,\mathrm{Im}\,G^R v_\beta\partial_\varepsilon G^A\big)\right], \tag{36}$$

where S is area of the system, $n_F(E-\mu) = 1/[e^{\beta(E-\mu)}+1]$ is the Fermi-Dirac distribution, $v = [r, H]/i\hbar$ is velocity operator, and the retarded/advanced Green function reads

$$G^{R/A} = \frac{1}{\varepsilon - H \pm i\delta} \tag{37}$$

with δ representing the disorder induced broadening, and in the clean limit $\delta \to 0^+$, we have

$$\mathrm{Im}\, G^R = -\pi\delta(\varepsilon - H). \tag{38}$$

Firstly we evaluate the Hall conductance with zero-magnetic field and zero-temperature, under which condition an equivalent formula for the considered two-band model[8] arises as

$$\sigma_{xy} = -\frac{e^2}{h}\frac{1}{4\pi}\int \Theta(\mu_F + d(k))\mathrm{d}^2k\frac{(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d}}{d^3}, \tag{39}$$

where the integral is down by partial integral over the occupied band bounded by Fermi energy. For the two band model, one recognizes $\boldsymbol{d}(k) = (v\hbar k_x, v\hbar k_y, M(\hbar^2k^2))$, and

$$\begin{aligned}(\partial_{k_x}\boldsymbol{d}\times\partial_{k_y}\boldsymbol{d})\cdot\boldsymbol{d} &= -v^2\hbar^2(k_x\partial_{k_x}M(\hbar^2k^2) + k_y\partial_{k_y}M(\hbar^2k^2) - M(\hbar^2k^2)) \\ &= -v^2\hbar^2(2k^2\partial_{k^2}M(\hbar^2k^2) - M(\hbar^2k^2)),\end{aligned} \tag{40}$$

and by defining

$$\cos\theta = \frac{M(\hbar^2k^2)}{[v^2\hbar^2 + M^2(\hbar^2k^2)]^{1/2}}, \tag{41}$$

we have

$$\partial_{k^2}\cos\theta = \frac{1}{2}\frac{v^2\hbar^2(2k^2\partial_{k^2}M(\hbar^2k^2) - M(\hbar^2k^2))}{[v^2\hbar^2 + M^2(\hbar^2k^2)]^{3/2}}, \tag{42}$$

so that

$$\begin{aligned}\sigma_{xy} &= \frac{e^2}{h}\frac{1}{2}\int_{k_F}^{+\infty}\mathrm{d}k^2\partial_{k^2}\cos\theta = \frac{e^2}{h}\frac{1}{2}[-\cos\theta(k_F) + \cos\theta(\infty)] \\ &= -\frac{e^2}{h}\frac{1}{2}\left[\mathrm{sgn}(b) + \frac{M(\hbar^2k_F^2)}{|\mu_F|}\right],\ \mu_F = -d(k_F),\end{aligned} \tag{43}$$

which shows explicitly a high-energy part contribution by $\mathrm{sgn}(b)$ with half-quantum-unit Hall contribution, and is a geometrical result with explicitly broken parity symmetry when $k \to \infty$. When $k_F < k_c$, we obtain an exactly half-quantized result $\sigma_{xy}/(e^2/h) = -\mathrm{sgn}(b)/2$, as a condensed matter analogy of quantum anomaly within frame of field theoretic description.

To evaluate Hall conductance under finite magnetic field, we need to carry out the represented velocity operator explicitly, which requires an obvious form of eigenstate wavefunction thus a gauge choice is needed. To simplify the calculation, here we choose the Landau gauge with $\boldsymbol{A} = (-By, 0, 0)$, under which condition p_x commutes with total Hamiltonian and remains translational invariance $p_x = \hbar k_x$ and

$$\begin{cases} a = [(y_0 - y)/l_B - l_B\partial_y]/\sqrt{2} \\ a^\dagger = [(y_0 - y)/l_B + l_B\partial_y]/\sqrt{2} \end{cases}, \tag{44}$$

the same as we defined before with $y_0 = k_x l_B^2$ the shifted center. Notice that since energy of state is independent of k_x, it offers us the Landau level degeneracy by

$$N_L = \frac{L_y}{\Delta y_0} = \frac{L_y}{l_B^2 \Delta k_x} = \frac{L_x L_y}{2\pi l_B^2} = S \cdot \frac{eB}{h}. \tag{45}$$

Now the eigenstate reads

$$|\varphi_{nsk_x}\rangle = \begin{pmatrix} c_{n1} |n-1, k_x\rangle \\ c_{n2} |n, k_x\rangle \end{pmatrix}, \ \varepsilon_{ns} = \frac{\omega_n}{2} + sE_n, \ E_n = \sqrt{m_n^2 + \eta^2 n}. \tag{46}$$

To get explicit wavefunction, notice that the deterministic equation reads

$$\begin{pmatrix} M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] - \varepsilon_{ns} & \eta\sqrt{n} \\ \eta\sqrt{n} & -M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right] - \varepsilon_{ns} \end{pmatrix} \begin{pmatrix} c_{n1} \\ c_{n2} \end{pmatrix} = 0,$$

$$\Longrightarrow \begin{pmatrix} m_n - sE_n & \eta\sqrt{n} \\ \eta\sqrt{n} & -m_n - sE_n \end{pmatrix} \begin{pmatrix} c_{n1} \\ c_{n2} \end{pmatrix} = sE_n \begin{pmatrix} \cos\theta_{ns} - 1 & s\sin\theta_{ns} \\ s\sin\theta_{ns} & -\cos\theta_{ns} - 1 \end{pmatrix} \begin{pmatrix} c_{n1} \\ c_{n2} \end{pmatrix} = 0,$$

where the sign-fixed angle is defined as

$$\begin{cases} \cos\theta_{ns} = sm_n/E_n \\ \sin\theta_{ns} = \eta\sqrt{n}/E_n \end{cases}. \tag{47}$$

Note that we have expelled the case $E_n = 0$ which happens only when $n = 0$ in our model with a straight solution $c_{01} = 0, \ c_{02} = 1$ illustrated above. Then by exploiting the first line, we obtain

$$(\cos\theta - 1)c_1 + s\sin\theta c_2 = 2\sin\frac{\theta}{2}(-\sin\frac{\theta}{2}c_1 + s\cos\frac{\theta}{2}c_2) = 0,$$

which leads to one real solution $(c_1, c_2) = (\cos\frac{\theta}{2}, s\sin\frac{\theta}{2})$ so that

$$|\varphi_{nsk_x}\rangle = \begin{pmatrix} \cos\frac{\theta_{ns}}{2} |n-1, k_x\rangle \\ s\sin\frac{\theta_{ns}}{2} |n, k_x\rangle \end{pmatrix}, \tag{48}$$

with its real-space wavefunction read as

$$\varphi_{nsk_x}(r) = \frac{e^{ik_x x}}{\sqrt{L_x l_B}} \begin{pmatrix} \cos\frac{\theta_{ns}}{2}\phi_{n-1}(\frac{y-y_0}{l_B}) \\ s\sin\frac{\theta_{ns}}{2}\phi_n(\frac{y-y_0}{l_B}) \end{pmatrix}, \tag{49}$$

with Hermite-Gaussian function defined above. Note that $m_{n<n_c} = 0$, and in the limit taking manner, the 0th coefficients read

$$\lim_{n\to 0^+} \cos\theta_0 = \lim_{n\to 0^+} \frac{m_n}{E_n} = 0,$$

$$\lim_{n\to 0^+} \sin\theta_0 = \lim_{n\to 0^+} \frac{\eta\sqrt{n}}{\sqrt{m_n^2 + \eta^2 n}} = 1,$$

then it is fairly-well to set $\theta_0 = \pi/2$ so that $\cos\theta_0/2 = \sin\theta_0/2 = \sqrt{2}/2$, which, however, making problem arise that the wavefunction is not well normalized, since as we have claimed before that $|c_{0,2}| = 1$ is needed. The compromise for this is to choose

$$\varphi_{0,-,k_x}(r) = \sqrt{2}\frac{e^{ik_x x}}{\sqrt{L_x l_B}} \begin{pmatrix} \cos\frac{\theta_0}{2}\phi_{-1}(\frac{y-y_0}{l_B}) \\ s\sin\frac{\theta_0}{2}\phi_0(\frac{y-y_0}{l_B}) \end{pmatrix}, \tag{50}$$

with $\sqrt{2}$ multiplying factor and defining $\phi_{-1} \equiv 0$.

Now to calculate the Hall conductance, firstly notice that under clean limit, the Kubo formula is reduced to

$$\sigma_{xy} = \mathrm{Im}\,\frac{e^2\hbar}{S}\sum_{\lambda,\lambda'}[n_F(\varepsilon_\lambda - \mu) - n_F(\varepsilon_{\lambda'} - \mu)]\frac{\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}}{(\varepsilon_\lambda - \varepsilon_{\lambda'})^2}, \tag{51}$$

with λ, λ' quantum number running over all possible values ($\lambda = \{n, s, k_x\}$ in the case), the key becomes evaluating velocity operator component

$$\langle v_\alpha\rangle_{\lambda\lambda'} = \frac{i}{\hbar}\langle\varphi_\lambda|\,[H, r_\alpha]\,|\varphi_{\lambda'}\rangle = \frac{i}{\hbar}(\varepsilon_\lambda - \varepsilon_{\lambda'})\langle\varphi_\lambda|r_\alpha|\varphi_{\lambda'}\rangle\,, \tag{52}$$

where we have explicitly utilized the eigen-equation to avoid the possible commutation problem arising from general irregular mass term. For Hall conductance, both x and y components need calculation, and for x part, we insert complete relation to find

$$\begin{aligned}
\langle x\rangle_{\lambda\lambda'} &= \left(c_{\lambda 1}\langle n-1, k_x|\ \ c_{\lambda 2}\langle n, k_x|\right)\int \mathrm{d}^2 r\,|r\rangle\langle r|\,x\int \mathrm{d}^2 r'\,|r'\rangle\langle r'|\begin{pmatrix} c_{\lambda' 1}|n'-1, k_x'\rangle \\ c_{\lambda' 2}|n', k_x'\rangle \end{pmatrix} \\
&= \frac{1}{L_x l_B}\int \mathrm{d}x x e^{-i(k_x - k_x')x} l_B(c_{\lambda 1}c_{\lambda' 1}\langle\phi_{n-1}, \phi_{n'-1}\rangle + c_{\lambda 2}c_{\lambda' 2}\langle\phi_n, \phi_{n'}\rangle) \\
&= \delta_{k_x, k_x'}(-i\partial_{k_x})(c_{\lambda 1}c_{\lambda' 1}\langle\phi_{n-1}, \phi_{n'-1}\rangle + c_{\lambda 2}c_{\lambda' 2}\langle\phi_n, \phi_{n'}\rangle) \\
&= i\delta_{k_x, k_x'} l_B(c_{\lambda 1}c_{\lambda' 1}\langle\phi'_{n-1}, \phi_{n'-1}\rangle + c_{\lambda 2}c_{\lambda' 2}\langle\phi'_n, \phi_{n'}\rangle) \\
&= i\delta_{k_x, k_x'} l_B\left[c_{\lambda 1}c_{\lambda' 1}\left(\sqrt{\frac{n}{2}}\delta_{n,n'-1} - \sqrt{\frac{n-1}{2}}\delta_{n-1,n'}\right) + c_{\lambda 2}c_{\lambda' 2}\left(\sqrt{\frac{n+1}{2}}\delta_{n+1,n'} - \sqrt{\frac{n}{2}}\delta_{n-1,n'}\right)\right] \\
&= i\delta_{k_x, k_x'} l_B\left[-\left(c_{\lambda 1}c_{\lambda' 1}\sqrt{\frac{n-1}{2}} + c_{\lambda 2}c_{\lambda' 2}\sqrt{\frac{n}{2}}\right)\delta_{n-1,n'} + \left(c_{\lambda 1}c_{\lambda' 1}\sqrt{\frac{n}{2}} + c_{\lambda 2}c_{\lambda' 2}\sqrt{\frac{n+1}{2}}\right)\delta_{n+1,n'}\right],
\end{aligned}$$

while for y, since $y = y_0 - (a + a^\dagger) l_B/\sqrt{2}$

$$\begin{aligned}
\langle y \rangle_{\lambda'\lambda} &= \left(c_{\lambda' 1} \langle n'-1, k_x'| \;\; c_{\lambda' 2} \langle n', k_x'| \right) \left(y_0 - l_B \frac{a + a^\dagger}{\sqrt{2}} \right) \begin{pmatrix} c_{\lambda 1} |n-1, k_x\rangle \\ c_{\lambda 2} |n, k_x\rangle \end{pmatrix} \\
&= \delta_{k_x, k_x'} y_0 (c_{\lambda' 1} c_{\lambda 1} + c_{\lambda' 2} c_{\lambda 2}) \delta_{nn'} \text{ (abandoned)} \\
&- \delta_{k_x, k_x'} \frac{l_B}{\sqrt{2}} \left(c_{\lambda' 1} \langle n'-1, k_x'| \;\; c_{\lambda' 2} \langle n', k_x'| \right) \begin{pmatrix} c_{\lambda 1} \sqrt{n-1} \, |n-2\rangle + c_{\lambda 1} \sqrt{n} \, |n\rangle \\ c_{\lambda 2} \sqrt{n} \, |n-1\rangle + c_{\lambda 2} \sqrt{n+1} \, |n+1\rangle \end{pmatrix} \\
&= -\delta_{k_x, k_x'} l_B \left[\left(c_{\lambda' 1} c_{\lambda 1} \sqrt{\frac{n-1}{2}} + c_{\lambda' 2} c_{\lambda 2} \sqrt{\frac{n}{2}} \right) \delta_{n-1, n'} + \left(c_{\lambda' 1} c_{\lambda 1} \sqrt{\frac{n}{2}} + c_{\lambda' 2} c_{\lambda 2} \sqrt{\frac{n+1}{2}} \right) \delta_{n+1, n'} \right],
\end{aligned}$$

where $n = n'$ part is thrown since it does not contribute to the Hall conductance counting on form of $\langle x \rangle$, which only contains nearest-neighbour contribution. Now by the same δ-pairing principle

$$\begin{aligned}
\langle v_x \rangle_{\lambda\lambda'} \langle v_y \rangle_{\lambda'\lambda} &= \frac{(\varepsilon_\lambda - \varepsilon_{\lambda'})^2}{\hbar^2} \langle x \rangle_{\lambda\lambda'} \langle y \rangle_{\lambda'\lambda} = \frac{i l_B^2 \delta_{k_x, k_x'}}{\hbar^2} (\varepsilon_\lambda - \varepsilon_{\lambda'})^2 \\
&\cdot \left[\left(c_{\lambda 1} c_{\lambda' 1} \sqrt{\frac{n-1}{2}} + c_{\lambda 2} c_{\lambda' 2} \sqrt{\frac{n}{2}} \right)^2 \delta_{n-1, n'} - \left(c_{\lambda 1} c_{\lambda' 1} \sqrt{\frac{n}{2}} + c_{\lambda 2} c_{\lambda' 2} \sqrt{\frac{n+1}{2}} \right)^2 \delta_{n+1, n'} \right] \\
&= \delta_{k_x, k_x'} [-i (v_{\lambda\lambda'})^2 \delta_{n+1, n'} + i (v_{\lambda'\lambda})^2 \delta_{n, n'+1}],
\end{aligned}$$

with definition

$$v_{\lambda\lambda'} \equiv \frac{l_B}{\sqrt{2}\hbar} (\varepsilon_\lambda - \varepsilon_{\lambda'}) (c_{\lambda 1} c_{\lambda' 1} \sqrt{n} + c_{\lambda 2} c_{\lambda' 2} \sqrt{n+1}), \tag{53}$$

which offers a general frame evaluating the single Dirac-like fermion under magnetic field. Notice that the formula is also suitable for $n = 0$ case, i.e.,

$$v_{0-,\lambda'} \equiv \frac{l_B}{\sqrt{2}\hbar} (\varepsilon_0 - \varepsilon_{\lambda'}) c_{0-,2} c_{\lambda' 2}, \tag{54}$$

but one needs to remember that $c_{0-,2} = -\sqrt{2}$ so to cancel the denominator.

Turning to the surface state model, firstly we notice that the deduced formula is self-adapted to $n = 0$ case by multiplying a $\sqrt{2}$ factor, i.e.,

$$\begin{aligned}
\langle x \rangle_{0-k_x, \lambda'} &= i \delta_{k_x, k_x'} l_B c_{0-,2} c_{\lambda' 2} \delta_{1, n'}, \\
\langle y \rangle_{\lambda', 0-k_x} &= -\delta_{k_x, k_x'} l_B c_{0-,2} c_{\lambda' 2} \delta_{1, n'}, \\
v_{0-,\lambda'} &= \frac{l_B}{\hbar} (\varepsilon_0 - \varepsilon_{\lambda'}) c_{0-,2} c_{\lambda' 2}
\end{aligned}$$

with $c_{0-,2} = \sqrt{2}s\sin\theta_0/2\big|_{s=-1} = -1$. And then we count on

$$\begin{aligned}
\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns}-\varepsilon_{n+1,s'}}\right]^2 &= (c_{ns,1}c_{n+1s',1}\sqrt{n}+c_{ns,2}c_{n+1s',2}\sqrt{n+1})^2 \\
&= \left[\cos\frac{\theta_{ns}}{2}\cos\frac{\theta_{n+1,s'}}{2}\sqrt{n}+ss'\sin\frac{\theta_{ns}}{2}\sin\frac{\theta_{n+1,s'}}{2}\sqrt{n+1}\right]^2 \\
&= \frac{1}{4}[(1+\cos\theta_{ns})(1+\cos\theta_{n+1,s'})n+(1-\cos\theta_{ns})(1-\cos\theta_{n+1,s'})(n+1)] \\
&+ \frac{ss'}{2}\sin\theta_{ns}\sin\theta_{n+1,s'}\sqrt{n(n+1)} \\
&= \frac{1}{4}\left[(2n+1)\left(1+ss'\frac{m_nm_{n+1}}{E_nE_{n+1}}\right)-s\frac{m_n}{E_n}-s'\frac{m_{n+1}}{E_{n+1}}\right]+\frac{ss'}{2}\frac{\eta^2n(n+1)}{E_nE_{n+1}},
\end{aligned}$$

which leads to

$$\sum_{s'}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns}-\varepsilon_{n+1,s'}}\right]^2 = \frac{1}{2}\left[2n+1-s\frac{m_n}{E_n}\right], \tag{55a}$$

$$\sum_{s}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns}-\varepsilon_{n+1,s'}}\right]^2 = \frac{1}{2}\left[2n+1-s'\frac{m_{n+1}}{E_{n+1}}\right]. \tag{55b}$$

However, notice that for $n=0$ case, $c_{0-,2} = -\sqrt{2}$ leads to

$$\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{0s,1s'}}{\varepsilon_{0,s}-\varepsilon_{1,s'}}\right]^2 = 2\times\frac{1}{4}\left[1-s'\frac{m_1}{E_1}\right] = \frac{1}{2}\left[1-s'\frac{m_1}{E_1}\right], \tag{56}$$

which further leads to

$$\sum_{s}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{0s,1s'}}{\varepsilon_{0,s}-\varepsilon_{1,s'}}\right]^2 = \frac{1}{2}\left[1-s'\frac{m_1}{E_1}\right], \tag{57a}$$

$$\sum_{s'}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{0s,1s'}}{\varepsilon_{0,s}-\varepsilon_{1,s'}}\right]^2 = 1, \tag{57b}$$

and one needs to be extremely carefully about this.

Now we evaluate the Hall conductance, which reads

$$
\begin{aligned}
\sigma_{xy}(B) &= \mathrm{Im}\,\frac{e^2\hbar}{S}\sum_{\lambda,\lambda'}[n_F(\varepsilon_\lambda-\mu)-n_F(\varepsilon_{\lambda'}-\mu)]\frac{\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}}{(\varepsilon_\lambda-\varepsilon_{\lambda'})^2}\\
&= \frac{e^2\hbar}{S}N_L\,\mathrm{Im}\sum_{ns,n's'}[n_F(\varepsilon_{ns}-\mu)-n_F(\varepsilon_{n's'}-\mu)]\frac{\langle v_x\rangle_{ns,n's'}\langle v_y\rangle_{n's',ns}}{(\varepsilon_{ns}-\varepsilon_{n's'})^2}\\
&= \frac{e^2\hbar}{2\pi l_B^2}\,\mathrm{Im}\sum_{ns,n's'}[n_F(\varepsilon_{ns}-\mu)-n_F(\varepsilon_{n's'}-\mu)]\frac{-i(v_{ns,n's'})^2\delta_{n+1,n'}+i(v_{n's',ns})^2\delta_{n,n'+1}}{(\varepsilon_{ns}-\varepsilon_{n's'})^2}\\
&= -\frac{e^2\hbar}{2\pi l_B^2}\sum_{nss'}[n_F(\varepsilon_{ns}-\mu)-n_F(\varepsilon_{n+1,s'}-\mu)]\frac{(v_{ns,n+1s'})^2}{(\varepsilon_{ns}-\varepsilon_{n+1,s'})^2}\\
&\quad+\frac{e^2\hbar}{2\pi l_B^2}\sum_{n'ss'}[n_F(\varepsilon_{n'+1,s}-\mu)-n_F(\varepsilon_{n's'}-\mu)]\frac{(v_{n's',n'+1s})^2}{(\varepsilon_{n'+1,s}-\varepsilon_{n',s'})^2}\\
&= -\frac{e^2\hbar}{2\pi l_B^2}\frac{l_B^2}{2\hbar^2}2\sum_n\left[\sum_s n_F(\varepsilon_{ns}-\mu)\sum_{s'}-\sum_{s'}n_F(\varepsilon_{n+1,s'}-\mu)\sum_s\right]\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns}-\varepsilon_{n+1,s'}}\right]^2\\
&= -\frac{e^2}{h}\sum_{s=-}n_F(\varepsilon_{0s}-\mu)+\frac{e^2}{h}\frac{1}{2}\sum_s n_F(\varepsilon_{1,s}-\mu)\left(1-s\frac{m_1}{E_1}\right)\\
&\quad-\frac{e^2}{h}\frac{1}{2}\lim_{N\to+\infty}\sum_{n=1}^{N}\sum_s\left[n_F(\varepsilon_{ns}-\mu)\left(2n+1-s\frac{m_n}{E_n}\right)-n_F(\varepsilon_{n+1,s}-\mu)\left(2n+1-s\frac{m_{n+1}}{E_{n+1}}\right)\right]\\
&= -\frac{e^2}{h}n_F(\varepsilon_0-\mu)+\frac{e^2}{h}\lim_{N\to+\infty}\frac{1}{2}\sum_s n_F(\varepsilon_{N+1,s}-\mu)\left(2N+1-s\frac{m_{N+1}}{E_{N+1}}\right)\\
&\quad-\frac{1}{2}\frac{e^2}{h}\lim_{N\to+\infty}\sum_{n=1}^{N}\sum_s\left[n_F(\varepsilon_{ns}-\mu)\left(2n+1-s\frac{m_n}{E_n}\right)-n_F(\varepsilon_{n,s}-\mu)\left(2n-1-s\frac{m_n}{E_n}\right)\right],
\end{aligned}
$$

and now by assuming that N is large enough so that $\varepsilon_{N+1,s}\to s\infty$, and we recognize that $n_F(\varepsilon_{N+1,s}-\mu)=0$ for $s=+$ and $=1$ for $s=-$, then we continue to have

$$
\begin{aligned}
\frac{\sigma_{xy}}{e^2/\hbar} &= -n_F(\varepsilon_0-\mu)+\lim_{N\to+\infty}\left[N+\frac{1}{2}+\frac{m_{N+1}}{2E_{N+1}}-\sum_{n=1}^{N}\sum_s n_F(\varepsilon_{ns}-\mu)\right]\\
&= -n_F(\varepsilon_0-\mu)+\lim_{N\to+\infty}\left(\frac{1}{2}+\frac{m_{N+1}}{2E_{N+1}}\right)-\lim_{N\to+\infty}\sum_{n=1}^{N}[n_F(\varepsilon_{n+}-\mu)+n_F(\varepsilon_{n-}-\mu)-1]\\
&= -n_F(\varepsilon_0-\mu)+\frac{1}{2}-\frac{1}{2}\mathrm{sgn}(b)-\sum_{n=1}^{+\infty}[n_F(\varepsilon_{n+}-\mu)-n_F(-\varepsilon_{n-}+\mu)],
\end{aligned}
$$

where we have utilized that

$$
\lim_{N\to+\infty}\frac{m_{N+1}}{E_{N+1}}=\lim_{n\to+\infty}\frac{m_n}{\sqrt{m_n^2+\eta^2 n}}=\lim_{n\to+\infty}\mathrm{sgn}(m_n)=-\mathrm{sgn}(b),
$$

and

$$
n_F(x)-1=\frac{1}{e^{\beta x}+1}-1=-\frac{1}{e^{-x}+1}=-n_F(-x).
$$

Re-write Hall conductance and we get

$$\sigma_{xy} = -\frac{e^2}{2h}\mathrm{sgn}(b) + \sigma_{xy}^0, \tag{58}$$

where the unregulated Hall conductance for Dirac fermion, by subtracting the half-quantized contribution from b-mass term, reads

$$\sigma_{xy}^0 = \frac{e^2}{2h}\left[1 - 2n_F(\varepsilon_0 - \mu) - 2\sum_{n\geq 1,s} s n_F(s\varepsilon_{ns} - s\mu)\right], \tag{59}$$

which is a half-quantized quantity for fully occupied case under low temperature. Notice that $-\mathrm{sgn}(b)/2$ shows a half-quantized Hall conductance coming from the $N \to +\infty$ contribution, i.e., the high-energy part contribution. Geometrically, $N \to +\infty$ corresponds to the $k_y \to +\infty$ as a rough correspondence in the continuum model, which again reveals the contribution away from the Dirac point. Also notice that the form $n_F(\varepsilon_{n+} - \mu) - n_F(-\varepsilon_{n-} + \mu)$ is rather reasonable, which describes zero Hall contribution when μ lies between $\varepsilon_{1,\pm}$, while being shift by unit when either upper or lower crossing one Landau level.

Aside from Hall conductance, we could also calculate longitudinal conductivity. Also start from the Kubo formula, by preserving δ-broadening,

$$\begin{aligned}
\sigma_{xx} &= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_k \int n_F(\epsilon-\mu)\mathrm{d}\epsilon \mathrm{Tr}\left[v_x\partial_\epsilon G^R v_x \,\mathrm{Im}\, G^R - v_x \,\mathrm{Im}\, G^R v_x \partial_\epsilon G^A\right] \\
&= \mathrm{Im}\,\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int n_F(\epsilon-\mu)\mathrm{d}\epsilon \left[\langle v_x\rangle_{mn}\partial_\epsilon G_n^R \langle v_x\rangle_{nm}\,\mathrm{Im}\, G_m^R - \langle v_x\rangle_{mn}\,\mathrm{Im}\, G_n^R \langle v_x\rangle_{nm}\partial_\epsilon G_m^A\right] \\
&\overset{\text{I.B.P}}{=} \frac{1}{2}\frac{e^2\hbar}{\pi S}\sum_{k,mn} \int -\partial_\epsilon n_F(\epsilon-\mu)\mathrm{d}\epsilon \left[\langle v_x\rangle_{mn}\,\mathrm{Im}\, G_n^R \langle v_x\rangle_{nm}\,\mathrm{Im}\, G_m^R - \langle v_x\rangle_{mn}\,\mathrm{Im}\, G_n^R \langle v_x\rangle_{nm}\,\mathrm{Im}\, G_m^A\right] \\
&= \frac{e^2\hbar}{\pi S}\sum_{k,mn} \int -\partial_\epsilon n_F(\epsilon-\mu)\mathrm{d}\epsilon |\langle v_x\rangle_{mn}|^2 \,\mathrm{Im}\, G_m^R \,\mathrm{Im}\, G_n^R.
\end{aligned}$$

Firstly we could consider an ideal Dirac cone at zero-temperature and zero chemical potential, whose Hamiltonian and eigen-quantities are

$$H = \lambda(k_x\sigma_x + k_y\sigma_y),\ E_\pm = \pm\lambda k,\ |\pm\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm e^{i\phi}\end{pmatrix},\ \tan\phi = \frac{k_y}{k_x},$$

which leads to the DC longitudinal conductivity

$$\begin{aligned}
\sigma_{xx} &\overset{T=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn}\int \delta(\epsilon-\mu)d\epsilon|\left\langle v_x\right\rangle_{mn}|^2 \operatorname{Im} G_m^R \operatorname{Im} G_n^R \\
&\overset{\mu=0}{=} \frac{e^2\hbar}{\pi S}\sum_{k,mn}\frac{\delta^2|\langle v_x\rangle_{mn}|^2}{(\epsilon_n^2+\delta^2)(\epsilon_m^2+\delta^2)} \\
&= 2\frac{e^2\hbar}{\pi S}\frac{S}{(2\pi)^2}\int d^2k\left[\frac{\delta^2|\langle v_x\rangle_{++}|^2}{(\lambda^2k^2+\delta^2)^2}+\frac{\delta^2|\langle v_x\rangle_{+-}|^2}{(\lambda^2k^2+\delta^2)^2}\right] \\
&= \frac{e^2}{2\pi^3\hbar}\int d^2k\frac{\lambda^2\delta^2}{(\lambda^2k^2+\delta^2)^2} \\
&\overset{t=(\lambda k/\delta)^2}{=} \frac{e^2}{h}\frac{1}{\pi}\int_0^{+\infty}dt\frac{1}{(t+1)^2} = \frac{e^2}{h}\frac{1}{\pi},
\end{aligned}$$

which shows minimal conductivity as quantum unit conductance over π. Now for the same question under finite magnetic field, assuming that Dirac cone is ideal (equivalent to push $k_c \to +\infty$, since we only care about low energy excitations), and $\varepsilon_{ns} = s\eta\sqrt{n}$ with $c_{\lambda 1} = \sqrt{2}/2,\ c_{\lambda 2} = s\sqrt{2}/2$, we have

$$\begin{aligned}
\sigma_{xx} &= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \delta(\epsilon)\mathrm{d}\epsilon|\left\langle v_x\right\rangle_{\lambda\lambda'}|^2 \operatorname{Im} G_\lambda^R \operatorname{Im} G_{\lambda'}^R \\
&= \frac{e^2\hbar}{\pi S}\sum_{nsk_x,n's'k_x'}\frac{(\varepsilon_{ns}-\varepsilon_{n's'})^2}{\hbar^2}\frac{|\left\langle x\right\rangle_{ns,n's'}|^2\delta^2}{(\varepsilon_{ns}^2+\delta^2)(\varepsilon_{n's'}^2+\delta^2)} \\
&= \frac{e^2}{\pi\hbar S}N_L l_B^2 2\sum_{nss'}\frac{\delta^2(\varepsilon_{ns}-\varepsilon_{n+1,s'})^2}{(\varepsilon_{ns}^2+\delta^2)(\varepsilon_{n+1,s'}^2+\delta^2)}\left(c_{ns,1}c_{n+1s',1}\sqrt{\frac{n}{2}}+c_{ns,2}c_{n+1s',2}\sqrt{\frac{n+1}{2}}\right)^2 \\
&= \frac{e^2}{h}\frac{2}{\pi}\sum_{nss'}\frac{\delta^2\eta^2(\sqrt{n}-ss'\sqrt{n+1})^2}{(\eta^2n+\delta^2)(\eta^2(n+1)+\delta^2)}\frac{1}{8}(\sqrt{n}+ss'\sqrt{n+1})^2 \\
&= \frac{e^2}{h}\frac{1}{\pi}\sum_n\frac{\delta^2\eta^2}{(\eta^2n+\delta^2)(\eta^2(n+1)+\delta^2)} \\
&= \frac{e^2}{h}\frac{\delta^2}{\pi}\sum_{n=0}\left[\frac{1}{\eta^2n+\delta^2}-\frac{1}{\eta^2(n+1)+\delta^2}\right] \\
&= \frac{1}{\pi}\frac{e^2}{h},
\end{aligned}$$

which again shows minimal conductivity.

When one wants to perform generically the numerical calculation of Hall conductance under finite temperature and disorder broadening, one needs to start from the original

Kubo formula

$$\begin{aligned}\sigma_{xy} &= \operatorname{Im}\frac{e^2\hbar}{\pi S}\sum_{\boldsymbol{k}}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\left[\operatorname{Tr}\left(v_x\partial_\varepsilon G^R v_y \operatorname{Im} G^R\right)-\operatorname{Tr}\left(v_x \operatorname{Im} G^R v_y \partial_\varepsilon G^A\right)\right]\\ &= \operatorname{Im}\frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\left\langle v_x\right\rangle_{\lambda\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}\left[\partial_\varepsilon G^R_{\lambda'}\operatorname{Im} G^R_\lambda - \operatorname{Im} G^R_{\lambda'}\partial_\varepsilon G^A_\lambda\right],\end{aligned}$$

where notice that from deduction above, $\left\langle v_x\right\rangle_{\lambda,\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}$ is purely imaginary so that

$$\begin{aligned}\sigma_{xy} &= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\operatorname{Im}\left(\left\langle v_x\right\rangle_{\lambda\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}\right)\left[\partial_\varepsilon \operatorname{Re} G^R_{\lambda'}\operatorname{Im} G^R_\lambda - \operatorname{Im} G^R_{\lambda'}\partial_\varepsilon \operatorname{Re} G^A_\lambda\right]\\ &= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\operatorname{Im}\left(\left\langle v_x\right\rangle_{\lambda\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}\right)\left[\frac{\delta[(\varepsilon-\varepsilon_{\lambda'})^2-\delta^2]}{[(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2]^2[(\varepsilon-\varepsilon_\lambda)^2+\delta^2]}-(\lambda\leftrightarrow\lambda')\right]\\ &= \frac{2e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\operatorname{Im}\left(\left\langle v_x\right\rangle_{\lambda\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}\right)\frac{\delta[(\varepsilon-\varepsilon_{\lambda'})^2-\delta^2]}{[(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2]^2[(\varepsilon-\varepsilon_\lambda)^2+\delta^2]},\end{aligned}$$

where relations

$$\begin{aligned}\partial_\varepsilon \operatorname{Re} G^R_{\lambda'} = \partial_\varepsilon\frac{\varepsilon-\varepsilon_{\lambda'}}{(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2} &= -\frac{(\varepsilon-\varepsilon_{\lambda'})^2-\delta^2}{[(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2]^2},\\ \operatorname{Im} G^R_\lambda &= -\frac{\delta}{(\varepsilon-\varepsilon_\lambda)^2+\delta^2},\\ \operatorname{Im}\left(\left\langle v_x\right\rangle_{\lambda,\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda}\right) &= -\operatorname{Im}\left(\left\langle v_x\right\rangle_{\lambda'\lambda}\left\langle v_y\right\rangle_{\lambda\lambda'}\right)\end{aligned}$$

have been used. Then by bringing

$$\left\langle v_x\right\rangle_{\lambda,\lambda'}\left\langle v_y\right\rangle_{\lambda'\lambda} = \delta_{k_x,k_x'}[-i(v_{\lambda\lambda'})^2\delta_{n+1,n'}+i(v_{\lambda'\lambda})^2\delta_{n,n'+1}]$$

inside the expression, one obtains that

$$\begin{aligned}\sigma_{xy} &= \frac{2e^2\hbar}{\pi S}N_L\sum_{ns,n's'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)[-(v_{\lambda\lambda'})^2\delta_{n+1,n'}+(v_{\lambda'\lambda})^2\delta_{n,n'+1}]\cdot\frac{\cdots}{\cdots}\\ &= \frac{e^2\hbar}{\pi^2 l_B^2}\lim_{N\to+\infty}\sum_{s,s'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\left\{\sum_{n=0}^{N}-(v_{ns,n+1s'})^2\frac{\delta[(\varepsilon-\varepsilon_{n+1,s'})^2-\delta^2]}{[(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]^2[(\varepsilon-\varepsilon_{ns})^2+\delta^2]}\right.\\ &\qquad\left.\sum_{n'=0}^{N}(v_{n's',n'+1s})^2\frac{\delta[(\varepsilon-\varepsilon_{n's'})^2-\delta^2]}{[(\varepsilon-\varepsilon_{n's'})^2+\delta^2]^2[(\varepsilon-\varepsilon_{n'+1,s})^2+\delta^2]}\right\}\\ &= \frac{2e^2\hbar}{\pi^2 l_B^2}\lim_{N,E\to+\infty}\sum_{n=0}^{N}\sum_{s,s'}\int_{-E}^{E}\mathrm{d}\varepsilon n_F(\varepsilon-\mu)(v_{ns,n+1s'})^2\frac{\delta^3[(\varepsilon-\varepsilon_{ns})^2-(\varepsilon-\varepsilon_{n+1,s'})^2]}{[(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]^2[(\varepsilon-\varepsilon_{ns})^2+\delta^2]^2}\\ &= \frac{e^2}{h}\frac{2}{\pi}\lim_{N,E\to+\infty}\sum_{n=0}^{N}\sum_{s,s'}\int_{-E}^{E}\mathrm{d}\varepsilon n_F(\varepsilon-\mu)\frac{\delta^3[(\varepsilon-\varepsilon_{ns})^2-(\varepsilon-\varepsilon_{n+1,s'})^2](\varepsilon_{ns}-\varepsilon_{n+1,s'})^2}{[(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]^2[(\varepsilon-\varepsilon_{ns})^2+\delta^2]^2}\\ &\qquad\cdot(c_{ns1}c_{n+1s'1}\sqrt{n}+c_{ns2}c_{n+1s'2}\sqrt{n+1})^2,\end{aligned}$$

which offers the starting point for numerical calculation. Notice that one needs to specify $n = 0$, $1 \leq n \leq n_c$ and $n_c + 1 \leq n \leq N$ as three different cases. However, notice that the integration includes contribution below the Fermi surface, which may cause convergence problem since in principle one needs to sum over all states below for convergence. From analysis above, it is better that we try to perform an integral-by-part (IBP) process to transit the derivative over energy variable ε to the Fermi-Dirac distribution n_F which offers a Lorentz-like distribution that approaches Delta-function, and this procedure efficiently transfers the whole integration nearly below Fermi surface to that of only near the Fermi surface, which not only eliminate convergence problem but also reflects actual transport process as a low energy physics.

Now we start from

$$\sigma_{xy} = \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\,\mathrm{Im}\left(\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}\right)\left[\partial_\varepsilon \,\mathrm{Re}\, G^R_{\lambda'}\,\mathrm{Im}\, G^R_\lambda - \mathrm{Im}\, G^R_{\lambda'}\partial_\varepsilon \,\mathrm{Re}\, G^A_\lambda\right],$$

where one finds that (say by *Mathematica*)

$$\begin{aligned}\partial_\varepsilon \,\mathrm{Re}\, G^R_{\lambda'}\,\mathrm{Im}\, G^R_\lambda - \mathrm{Im}\, G^R_{\lambda'}\partial_\varepsilon \,\mathrm{Re}\, G^A_\lambda &= \frac{\delta[(\varepsilon-\varepsilon_{\lambda'})^2-\delta^2]}{[(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2]^2[(\varepsilon-\varepsilon_\lambda)^2+\delta^2]} - (\lambda\leftrightarrow\lambda')\\ &= \frac{\delta}{(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2}\frac{2\delta^2}{[(\varepsilon-\varepsilon_\lambda)^2+\delta^2]^2} - (\lambda\leftrightarrow\lambda')\\ &= \partial_\varepsilon f_{\lambda'\lambda}(\varepsilon),\end{aligned}$$

with definition

$$f_{\lambda'\lambda}(\varepsilon) \equiv \frac{1}{(\varepsilon_{\lambda'}-\varepsilon_\lambda)^2}\left[\tan^{-1}\frac{\varepsilon_{\lambda'}-\varepsilon}{\delta} + \tan^{-1}\frac{\varepsilon-\varepsilon_\lambda}{\delta} - \frac{\delta(\varepsilon_{\lambda'}-\varepsilon_\lambda)[(\varepsilon_{\lambda'}-\varepsilon)(\varepsilon_\lambda-\varepsilon)+\delta^2]}{[(\varepsilon-\varepsilon_{\lambda'})^2+\delta^2][(\varepsilon-\varepsilon_\lambda)^2+\delta^2]}\right]. \tag{60}$$

This function then turns out to be extremely important when evaluating the Hall conductance.

Now

$$\begin{aligned}
\sigma_{xy} &= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon n_F(\varepsilon-\mu)\,\mathrm{Im}\left(\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}\right)\partial_\varepsilon f_{\lambda'\lambda}(\varepsilon)\\
&= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon[-\partial_\varepsilon n_F(\varepsilon-\mu)]\,\mathrm{Im}\left(\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}\right) f_{\lambda'\lambda}(\varepsilon)\\
&= \frac{e^2\hbar}{\pi S}N_L\sum_{ns,n's'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)[-(v_{\lambda\lambda'})^2\delta_{n+1,n'}+(v_{\lambda'\lambda})^2\delta_{n,n'+1}]f_{n's',ns}\\
&= \frac{e^2\hbar}{\pi^2 l_B^2}\sum_{nss'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)(v_{ns,n+1s'})^2 f_{ns,n+1s'}\\
&= \frac{e^2\hbar}{\pi^2 l_B^2}\frac{l_B^2}{2\hbar^2}\sum_{nss'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)(\varepsilon_{ns}-\varepsilon_{n+1,s'})^2 f_{ns,n+1s'}(c_{ns1}c_{n+1s'1}\sqrt{n}+c_{ns2}c_{n+1s'2}\sqrt{n+1})^2\\
&= \frac{e^2}{h}\frac{1}{\pi}\sum_{nss'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)\left[\tan^{-1}\frac{\varepsilon_{ns}-\varepsilon}{\delta}+\tan^{-1}\frac{\varepsilon-\varepsilon_{n+1,s'}}{\delta}\right.\\
&\quad\left.-\frac{\delta(\varepsilon_{ns}-\varepsilon_{n+1,s'})[(\varepsilon_{ns}-\varepsilon)(\varepsilon_{n+1,s'}-\varepsilon)+\delta^2]}{[(\varepsilon-\varepsilon_{ns})^2+\delta^2][(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]}\right](c_{ns1}c_{n+1s'1}\sqrt{n}+c_{ns2}c_{n+1s'2}\sqrt{n+1})^2,
\end{aligned}$$

which serves as the new starting point for numerical calculation, where one may have to be careful about

$$-\partial_\varepsilon n_F(x)=\frac{\beta e^{\beta x}}{(e^{\beta x}+1)^2}=\frac{\beta}{4[\cosh(\beta x/2)]^2},$$

inside which the latter expression is more safe to perform without overflow problem.

Although more realistic for actual calculation, formula here still bears problem like low convergence rate with increasing of n. As we have deduced above for the clean case, we see that the 'anomaly' part contribution comes from a limit taking $\lim_{n\to+\infty} m_n/E_n=-\mathrm{sgn}(b)$, and one could imagine that to achieve such a limit how many Landau levels are needed, e.g. as shown in Fig. 1, number of LLs is chosen to be $10n_c$ for each magnetic field strength, however, the anomalous Hall conductance contributed from high-energy part still does not converge to 0.5 quantum unit. In order to solve the problem, one has to further deduce the formula above. Notice that by $-\partial_\varepsilon n_F$ the integral kernel contributes only near $\varepsilon\sim\mu$ for sufficiently low temperature, then for ultra-high LLs, we could treat $|\varepsilon_{ns}|\gg|\mu|$ so that

$$f_{ns,n+1s'}(\varepsilon)=\tan^{-1}\frac{\varepsilon_{ns}-\varepsilon}{\delta}+\tan^{-1}\frac{\varepsilon-\varepsilon_{n+1,s'}}{\delta}=\tan^{-1}\frac{\varepsilon_{ns}}{\delta}-\tan^{-1}\frac{\varepsilon_{n+1,s'}}{\delta}=\frac{\pi}{2}(s-s'),\tag{61}$$

where we have also used that $|\varepsilon_{ns}|\gg\delta$ for large n. Notice that the thrown term bears ε^{-1}

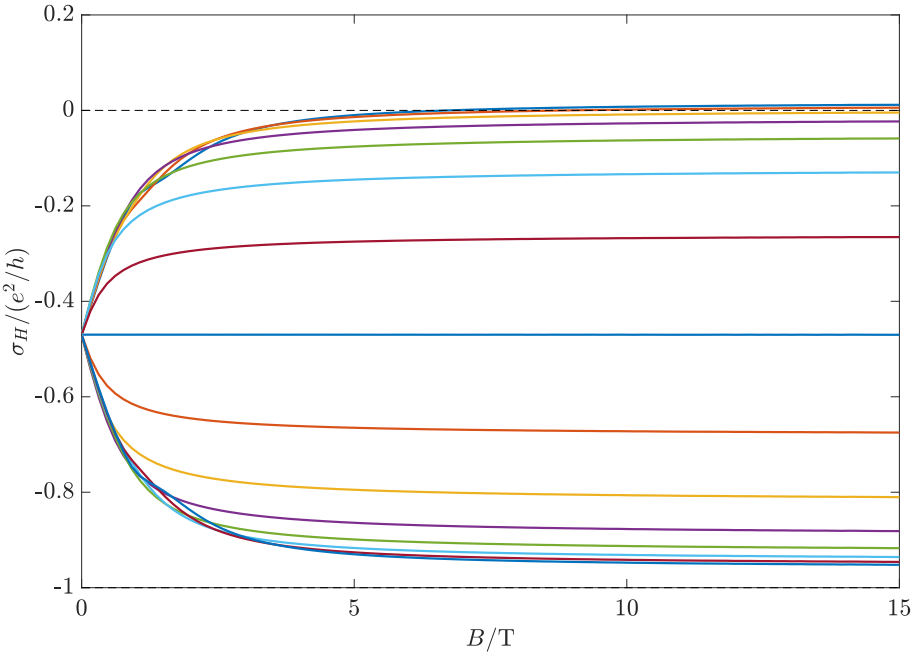


FIG. 1. Non-convergent Hall conductance with disorder-broadening $\delta = 10\text{meV}$, temperature $T = 1\text{K}$, and parameter set $v\hbar = 410\text{meV}\cdot\text{nm}$, $b\hbar^2 = 566\text{meV}\cdot\text{nm}^2$, $k_c = 0.7\text{nm}^{-1}$. The chemical has range $\mu \in [-25, 25]\text{meV}$.

scaling. With this simplification, the ε related integral is reduced to

$$\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F) = -n_F\Big|_{-\infty}^{+\infty} = 1.$$

Now we write

$$\begin{aligned}
\sigma_{xy} &= \sigma_{xy}(N) + \frac{e^2}{h}\frac{1}{\pi}\sum_{n=N+1}^{+\infty}\sum_{ss'}\frac{\pi}{2}(s-s')(c_{ns1}c_{n+1s'1}\sqrt{n} + c_{ns2}c_{n+1s'2}\sqrt{n+1})^2 \\
&= \sigma_{xy}(N) + \frac{e^2}{2h}\sum_{n=N+1}^{+\infty}\sum_{ss'}\frac{1}{4}\left(\frac{m_{n+1}}{E_{n+1}} - \frac{m_n}{E_n}\right) \\
&= \sigma_{xy}(N) + \frac{e^2}{2h}\sum_{n=N+1}^{+\infty}\left(\frac{m_{n+1}}{E_{n+1}} - \frac{m_n}{E_n}\right) \\
&= \sigma_{xy}(N) + \frac{e^2}{2h}\left(\lim_{n\to+\infty}\frac{m_n}{E_n} - \frac{m_{N+1}}{E_{N+1}}\right) \\
&= \sigma_{xy}(N) - \frac{e^2}{2h}\left(\mathrm{sgn}(b) + \frac{m_{N+1}}{E_{N+1}}\right),
\end{aligned}$$

where $N \gg n_c$ is the part to be calculated numerically, while the second term compensates for the 'anomaly' part deviation by previous N-summation. Result with new formula is

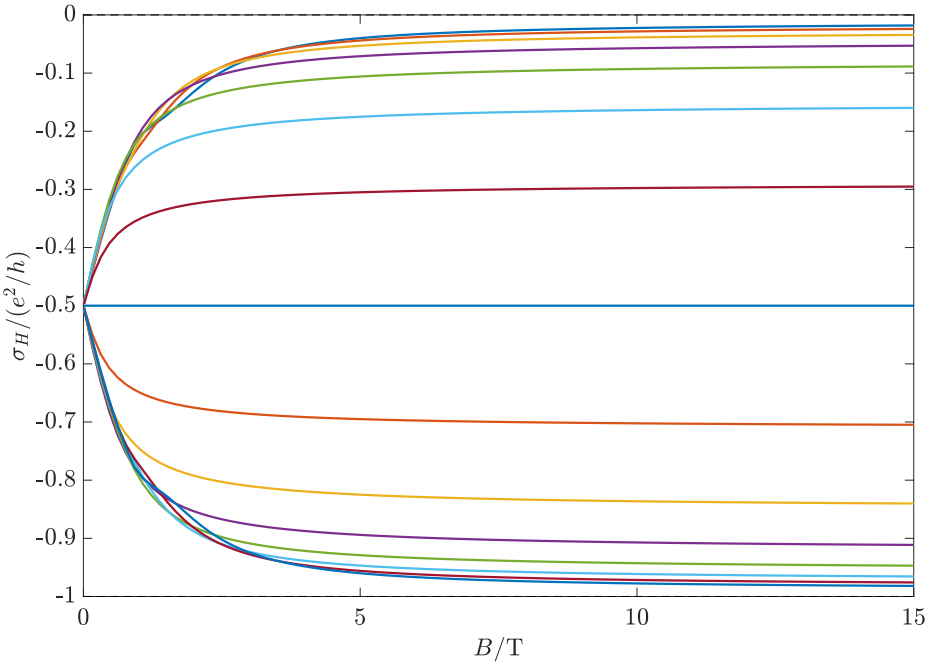


FIG. 2. Hall conductance with disorder-broadening $\delta = 10\text{meV}$, temperature $T = 1\text{K}$, and parameter set $v\hbar = 410\text{meV}\cdot\text{nm}$, $b\hbar^2 = 566\text{meV}\cdot\text{nm}^2$, $k_c = 0.7\text{nm}^{-1}$. The chemical has range $\mu \in [-25, 25]\text{meV}$.

shown in Fig. 2, a very attractive result is that since now disorder-broadening δ is larger or at least comparable with chemical potential μ, one could see the half-to-integer transition either by increasing the magnetic field (which suppresses broadening influence) or deviating μ from zero at sufficient B. Notice that for $\mu = 0$, the zeroth LL is always half-filling which leads to constant half-quantization. As should be put in the front, we can see how Hall conductance shows half-integer to integer transition at strong disorder with increasing B, as shown in Fig. 3.

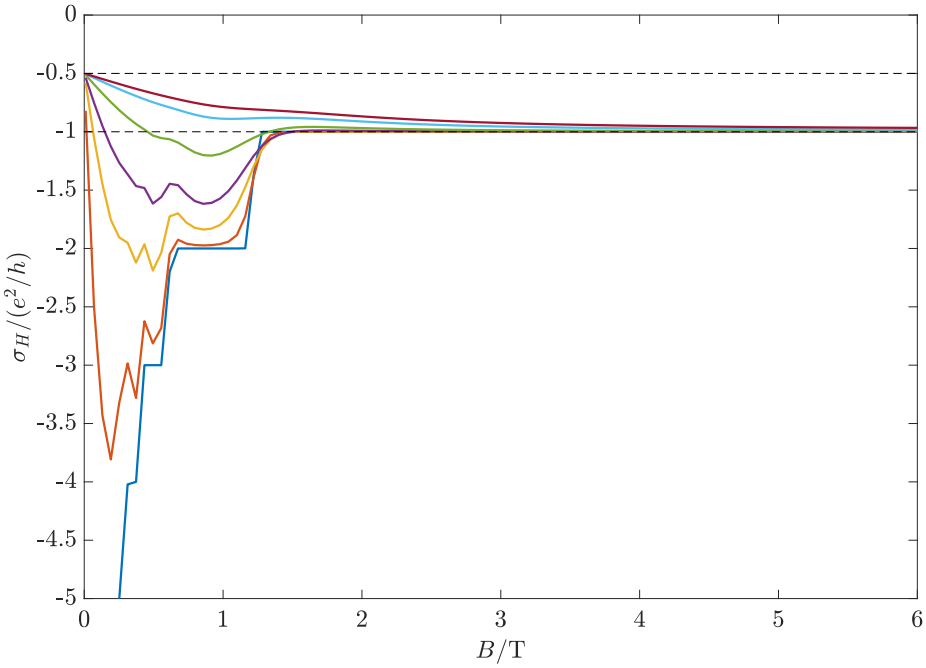


FIG. 3. Hall conductance with disorder-broadening $\delta = \{0.01, 1, 2, 3, 5, 8, 10\}$meV, temperature $T = 1$K, chemical potential $\mu = 25$meV, and parameter set $v\hbar = 410\text{meV}\cdot\text{nm}$, $b\hbar^2 = 566\text{meV}\cdot\text{nm}^2$, $k_c = 0.7\text{nm}^{-1}$.

The same procedure is true for longitudinal conductance, which reads

$$\begin{aligned}
\sigma_{xx} &= \frac{e^2\hbar}{\pi S}\sum_{\lambda,\lambda'}\int \mathrm{d}\varepsilon[-\partial_\varepsilon n_F(\varepsilon-\mu)]|\langle v_x\rangle_{\lambda\lambda'}|^2\,\mathrm{Im}\,G^R_\lambda\,\mathrm{Im}\,G^R_{\lambda'} \\
&= \frac{e^2\hbar}{\pi S}N_L\sum_{ns,n's'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)\frac{\delta^2[(v_{ns,n's'})^2\delta_{n+1,n'}+(v_{n's',ns})^2\delta_{n,n'+1}]}{[(\varepsilon-\varepsilon_{ns})^2+\delta^2][(\varepsilon-\varepsilon_{n's'})^2+\delta^2]} \\
&= \frac{e^2\hbar}{\pi l_B^2}\sum_{nss'}\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)\frac{\delta^2(v_{ns,n+1s'})^2}{[(\varepsilon-\varepsilon_{ns})^2+\delta^2][(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]} \\
&= \frac{e^2}{h}\frac{1}{\pi}\lim_{N,E\to+\infty}\sum_{n=0}^{N}\sum_{s,s'}\int_{-E}^{E}\mathrm{d}\varepsilon(-\partial_\varepsilon n_F)\frac{\delta^2(\varepsilon_{ns}-\varepsilon_{n+1,s'})^2}{[(\varepsilon-\varepsilon_{ns})^2+\delta^2][(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]} \\
&\qquad\cdot(c_{ns1}c_{n+1s'1}\sqrt{n}+c_{ns2}c_{n+1s'2}\sqrt{n+1})^2.
\end{aligned}$$

For actual calculation, one has to distinguish $n = 0$ case and others. There needs no large-n correction since they do not contribute to longitudinal conductance from the beginning for realistic chemical potential.

For numerical results, we firstly show, in Fig. 4, the Hall conductance in the clean limit, where for different chemical potential values the quantum Hall plateaus appear and so

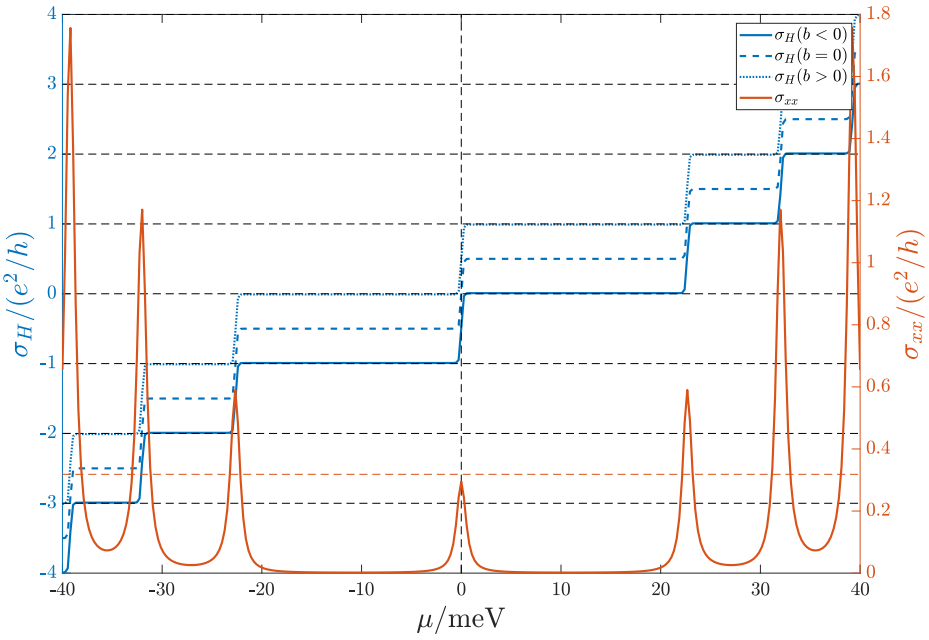


FIG. 4. (Left) Hall conductance in the clean limit. (Right) Longitudinal conductance. We choose $B = 1\mathrm{T}$ and $T = 1\mathrm{K}$, with parameter set $v\hbar = 410\mathrm{meV}\cdot\mathrm{nm}$, $b\hbar^2 = 566\mathrm{meV}\cdot\mathrm{nm}^2$, and $k_c = 0.7\mathrm{nm}^{-1}$[9]. Notice that Hall conductance is calculated with $b\hbar^2/\mathrm{meV}\cdot\mathrm{nm}^2 = \{-566, 0, 566\}$ separately, and finite disorder broadening $\delta = 0.5\mathrm{meV}$ has been chosen for longitudinal conductance calculation.

does their transitions, at which points the longitudinal conductance becomes non-vanishing. This is understood from the formula as $-\partial_\varepsilon n_F(\varepsilon-\mu) \sim \delta(\varepsilon-\mu)$, which then gives main contribution around μ, while the denominator suggests that only when $\varepsilon \sim \varepsilon_{ns}$ does the contribution becomes distinguishable, and totally, the analysis suggests that only when $\mu \sim \varepsilon_{ns}$ does longitudinal conductance appears. What is more, when the Fermi surface (FS) lies in the parity-invariant-regime (PIR), the longitudinal conductance shows approximately $\sigma_{xx}(n) \propto n$ scaling behaviour (not suitable for $n = 0$ by minimal conductance), since one could see from the integral that ε dependent terms give

$$\int \mathrm{d}\varepsilon(-\partial_\varepsilon n_F)\frac{\delta^2}{[(\varepsilon-\varepsilon_{ns})^2+\delta^2][(\varepsilon-\varepsilon_{n+1,s'})^2+\delta^2]},$$

where three Delta-like functions appear which leads to $\varepsilon_{ns} \sim \mu \sim \varepsilon_{n+1,s'}$, then only the cases $s = s'$ give considerable contribution, and this observation leads to that

$$\begin{aligned}&(\varepsilon_{ns}-\varepsilon_{n+1,s'})^2(c_{ns1}c_{n+1s'1}\sqrt{n}+c_{ns2}c_{n+1s'2}\sqrt{n+1})^2\\ \sim&\frac{\eta^2}{2}[(\sqrt{n}-\sqrt{n+1})(\sqrt{n}+\sqrt{n+1})]^2=\frac{\eta^2}{2},\end{aligned}$$

which only gives a nearly constant factor, and the varying contribution comes from the ε dependent integral. To evaluate its order, we can consider low-temperature limit so that $(-\partial_\varepsilon n_F) \sim \delta(\varepsilon - \mu)$, also notice that longitudinal peaks when μ lies at some Landau level reflected by near-Delta function nature, we could fairly-well set $\mu = \varepsilon_{ns}$ to see the change, which leads to

$$(s' = s) \int \mathrm{d}\varepsilon \delta(\varepsilon - \varepsilon_{ns}) \frac{\delta^2}{[(\varepsilon - \varepsilon_{ns})^2 + \delta^2][(\varepsilon - \varepsilon_{n+1,s'})^2 + \delta^2]} = \frac{1}{(\varepsilon_{ns} - \varepsilon_{n+1,s})^2 + \delta^2},$$

where the denominator contains factor

$$\sqrt{n+1} - \sqrt{n} = \sqrt{n}\left(1 + \frac{1}{2n} + O\left(\frac{1}{n^2}\right) - n\right) \approx \frac{1}{2\sqrt{n}},$$

which finally leads to the order-factor by ignoring disorder-broadening

$$\frac{1}{(\eta/2\sqrt{n})^2} = \frac{4n}{\eta^2},$$

which combined with the pre-deduced factor leads to

$$\sigma_{xx}(n) \sim \frac{e^2}{h}\frac{1}{\pi}\frac{\eta^2}{2}\frac{4n}{\eta^2} = 2n\frac{e^2}{\pi h},\ n \neq 0,\ n < n_c. \tag{62}$$

Reflected in the figure, since $\mu = \varepsilon_{ns} = \eta\sqrt{n}$ leads to $n = (\mu/\eta)^2$, then totally we should have

$$\sigma_{xx}(|\mu| < \varepsilon_{n_c,+}) \sim 2\left(\frac{\mu}{\eta}\right)^2 \frac{e^2}{\pi h}\delta_{\mu,\varepsilon_{ns}}, \tag{63}$$

with a quadratic power, as reflected in the figure. It is rather interesting that when the chemical potential stays away from the PIR, it would still preserve the n-scaling behavior, where by stay-away I mean the quadratic term dominates so that $\varepsilon_{ns} \sim s(n - n_c)\omega$. At this time, the integral on the contrary does not provide with scaling power, since

$$\frac{1}{(\varepsilon_{ns} - \varepsilon_{n+1,s})^2 + \delta^2} \sim \frac{1}{\omega^2},$$

while term

$$(\varepsilon_{ns} - \varepsilon_{n+1,s'})^2(c_{ns1}c_{n+1s'1}\sqrt{n} + c_{ns2}c_{n+1s'2}\sqrt{n+1})^2 \sim \omega^2\frac{1}{2}(2n+1) = n\omega^2,$$

and totally this yields that

$$\sigma_{xx}(n \gg n_c) \sim n\frac{e^2}{\pi h}, \tag{64}$$

which combined with $\mu = \varepsilon_{ns} \sim \omega(n - n_c)$ leads to

$$\sigma_{xx}(|\mu| \gg \varepsilon_{n_c,+}) \sim \left(\frac{\mu}{\omega} + n_c\right)\frac{e^2}{\pi h}. \tag{65}$$

The validity of parabolic dependence of $\sigma_{xx}(\mu)$ is presented in Fig. 5.

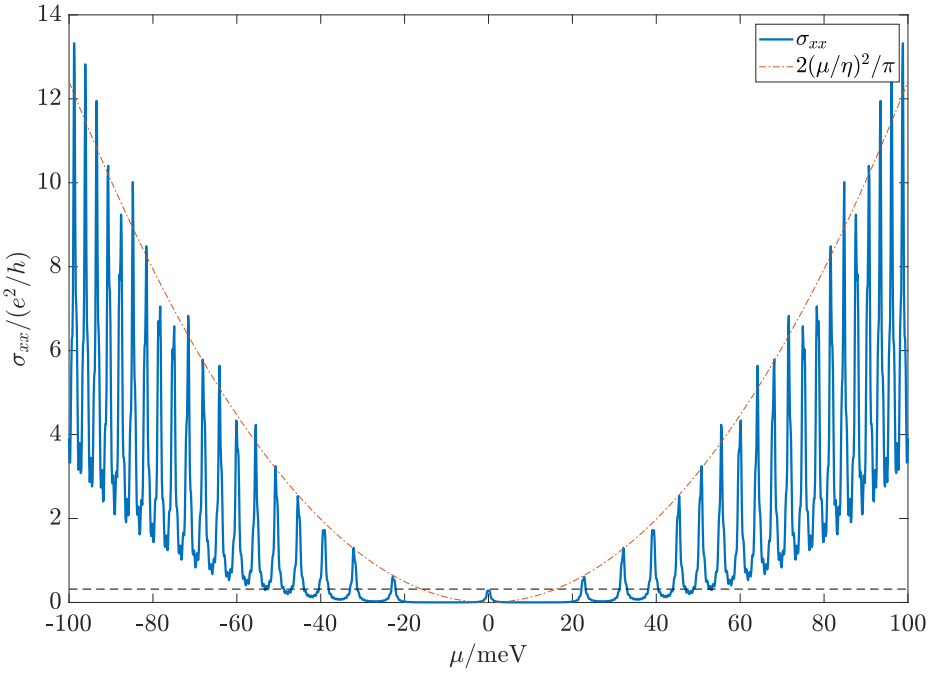


FIG. 5. σ_{xx} v.s. Eq. (63) as a parabolic dependence. We have chosen low temperature $T = 0.1$K with disorder-broadening $\delta = 0.5$meV.

III. Landau Level with TI Film

A film contains two surfaces thus two surface states, and the effective lowest four-band model could be written as[2]

$$H_S = \bigoplus_{\chi=\pm} h_\chi = \bigoplus_{\chi=\pm} \lambda[\sin(k_x a)\sigma_x + \sin(k_y a)\sigma_y] + \chi\Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k})\sigma_z, \tag{66}$$

under the mirror representation reflected by mirror eigenvalue χ. There are two ways to shift away two Dirac fermions presented.

• First way is to add z-asymmetric x-orbital-dependent field

$$V_O^x(l_z) = V^x(l_z)\sigma_0\tau_x,\ V^x(l_z) = -V^x(-l_z),$$

which will be projected into

$$V_s = V_A^x(k)\tau_z\sigma_0,$$

and thus shift two cones. Notice that in this case, the energy splitting happens in the mirror representation, i.e., we shift mirror-Dirac-fermions.

• Second way is to add z-asymmetric gate-voltage

$$V_G(l_z) = V(l_z)\sigma_0\tau_0,\ V(l_z) = -V(-l_z),$$

which will be projected into

$$V_A(k)\tau_y\sigma_0,$$

and by an accompanied unitary transformation on projecting basis which does not change form of H_S but bring it into surface-state-representation

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}\sigma_0,$$

we obtain

$$V_s = UV_A(k)\tau_y\sigma_0U^\dagger = V_A(k)\tau_z\sigma_0, \tag{67}$$

which splits the top and bottom surface states energy.

In any case, let us assume that projected potential is some constant independent of k, we obtain total Hamiltonian

$$H = H_S + V_s = \lambda[\sin(k_xa)\sigma_x + \sin(k_ya)\sigma_y] + \Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k})\tau_z\sigma_z + V\tau_z\sigma_0, \tag{68}$$

which is also separable as

$$H = \bigoplus_{\chi=\pm}\lambda[\sin(k_xa)\sigma_x + \sin(k_ya)\sigma_y] + \chi\left[\Theta(-m_0(\boldsymbol{k}))m_0(\boldsymbol{k})\sigma_z + V\sigma_0\right]. \tag{69}$$

Now let us consider its Landau-Level problem, and clearly, the changes are

$$M(\Pi^2), \omega_n, m_n \longrightarrow \chi M(\Pi^2), \chi\omega_n, \chi m_n, \tag{70a}$$

$$\sin\theta_{\chi ns}, \cos\theta_{\chi ns} \longrightarrow \sin\theta_{ns}, \chi\cos\theta_{ns}, \tag{70b}$$

$$\varepsilon_{ns} \longrightarrow \varepsilon_{\chi ns}(V) = \begin{cases} \chi V + s\eta\sqrt{n}, & n < n_c \\ \chi(\omega/2 + V) + s\sqrt{m_n^2 + \eta^2 n}, & n > n_c \end{cases}, \tag{70c}$$

which leads to

$$\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v^\chi_{ns,n+1s'}}{\varepsilon_{\chi ns} - \varepsilon_{\chi n+1,s'}}\right]^2 = \frac{1}{4}\left[(2n+1)\left(1 + ss'\frac{m_nm_{n+1}}{E_nE_{n+1}}\right) - s\chi\frac{m_n}{E_n} - s'\chi\frac{m_{n+1}}{E_{n+1}}\right] + \frac{ss'}{2}\frac{\eta^2 n(n+1)}{E_nE_{n+1}}. \tag{71}$$

This finally leads to, for Hall conductance under finite B, T, μ while being clean $\delta \to 0^+$,

$$\begin{aligned}\frac{\sigma_{xy}^{\chi}}{e^2/h} &= -n_F(\varepsilon_{\chi 0}-\mu) + \lim_{N\to+\infty}\left(\frac{1}{2}+\chi\frac{m_{N+1}}{2E_{N+1}}\right) - \lim_{N\to+\infty}\sum_{n=1}^{N}[n_F(\varepsilon_{\chi n+}-\mu)+n_F(\varepsilon_{\chi n-}-\mu)-1] \\ &= -\frac{\chi}{2}\mathrm{sgn}(b)+\sigma_{xy}^{0,\chi},\end{aligned} \tag{72}$$

where

$$\frac{\sigma_{xy}^{0,\chi}}{e^2/h} = \frac{1}{2} - n_F(\varepsilon_{\chi 0}-\mu) - \sum_{n\geq 1}\left[n_F(\varepsilon_{\chi n+}-\mu) - n_F(-\varepsilon_{\chi n-}+\mu)\right]. \tag{73}$$

And by adding contributions from two cones together, we find that the high-energy 'anomaly' contributed part $\chi\mathrm{sgn}(b)/2$ is exactly cancelled by each other, while the remaining part reads

$$\begin{aligned}\sigma_{xy}(V,B,T,\mu,\delta\to 0) &= \sigma_{xy}^{+}+\sigma_{xy}^{-} \\ &= \frac{e^2}{h}\left[1-\sum_{\chi} n_F(\varepsilon_{\chi 0}(V)-\mu) - \sum_{\chi,n\geq 1,s} s n_F(s\varepsilon_{\chi ns}(V)-s\mu)\right].\end{aligned} \tag{74}$$

Drawing Hall conductance leads to result in Fig. 6. We can also plot the Hall transition

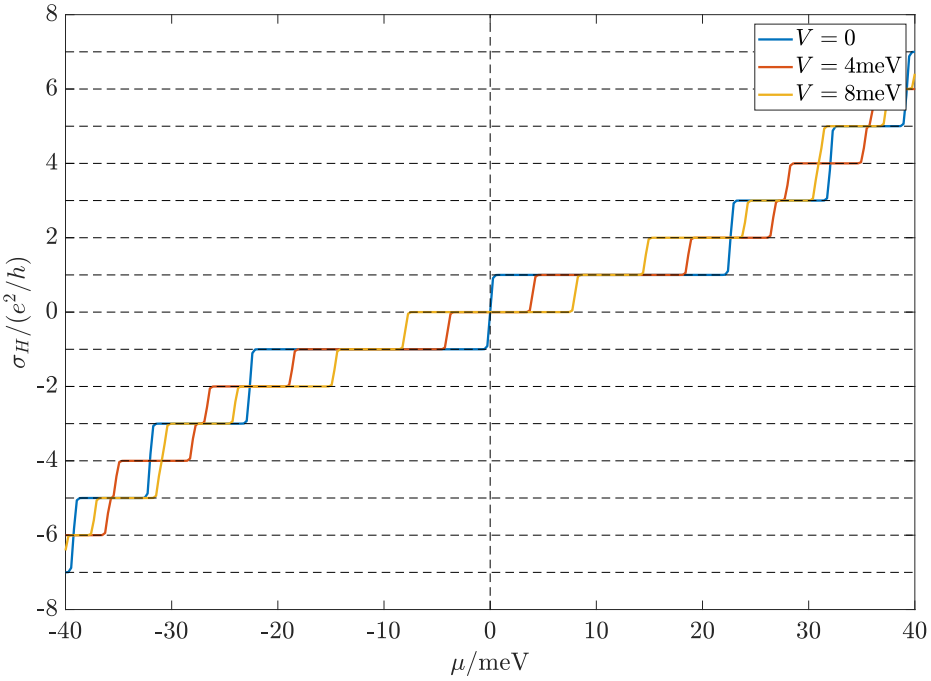


FIG. 6. Clean Hall conductance with temperature $T = 1$K, $B = 1$T, and parameter set $v\hbar = 410\text{meV}\cdot\text{nm}$, $b\hbar^2 = 566\text{meV}\cdot\text{nm}^2$, $k_c = 0.7\text{nm}^{-1}$, $V = \{0, 4, 8\}$meV. The chemical has range $\mu \in [-40, 40]$meV.

picture, as shown in Fig. 7, where the 'anomaly' is again cancelled, so does the gate-voltage asymmetry.

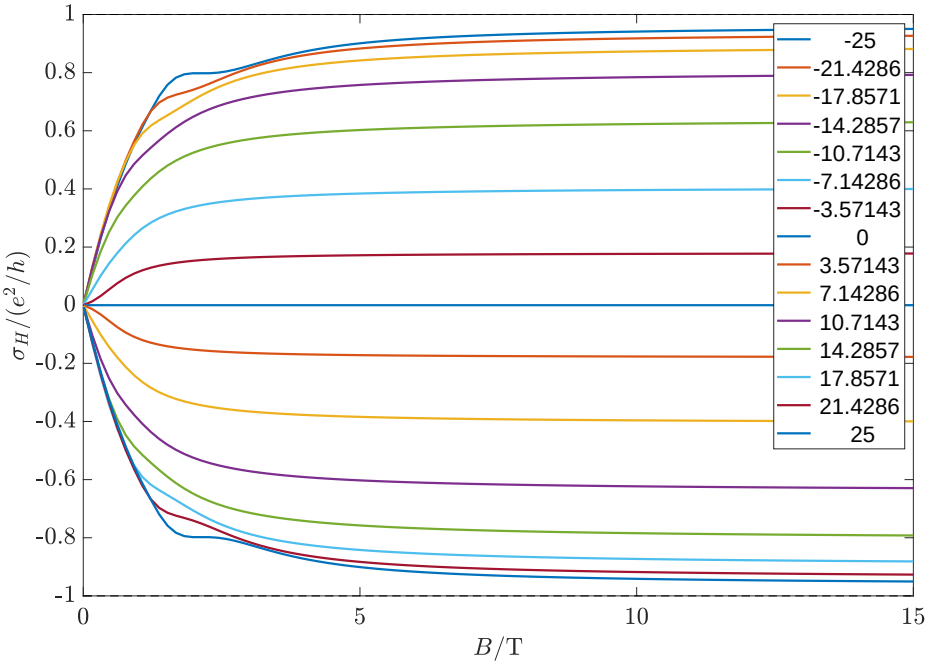


FIG. 7. Hall conductance with disorder-broadening $\delta = 10\text{meV}$, temperature $T = 1\text{K}$, gate-voltage asymmetry $V = 8\text{meV}$, and parameter set $v\hbar = 410\text{meV}\cdot\text{nm}$, $b\hbar^2 = 566\text{meV}\cdot\text{nm}^2$, $k_c = 0.7\text{nm}^{-1}$. The chemical has range $\mu \in [-25, 25]\text{meV}$.

Recently found that the gate-voltage term can not be simultaneously diagonalized with the surface Hamiltonian, so the above results are not fully correct.

IV. Landau Level with Opposite Magnetic Field

The above discussion is based on the assumption that the magnetic field is positive along z-direction. However, when the field direction is reversed, the constructing basis of Landau levels, i.e., the annihilation and creation operators, will be changed, which leads to the total modification of the solution structure and expression of Hall conductance.

By $\boldsymbol{B} = B\hat{z}$ $(B < 0)$, we need

$$a = \frac{l_B}{\sqrt{2}\hbar}(\Pi_x + i\Pi_y),\ a^\dagger = \frac{l_B}{\sqrt{2}\hbar}(\Pi_x - i\Pi_y), \tag{75}$$

such that $[a, a^\dagger] = 1$ is satisfied. Notice that we may also reverse the sign of a defined in $B > 0$ case to satisfy the commutation relation, which, however, will lead to unnormalizable

wavefunctions. Under position representation, we can write

$$a = \frac{1}{\sqrt{2}}[(y+y_0)/l_B + l_B\partial_y],\ a^\dagger = \frac{1}{\sqrt{2}}[(y+y_0)/l_B - l_B\partial_y], \tag{76}$$

and the wavefunction will be related to Hermite-Gaussian function with variable $(y+y_0)/l_B$.

For a single Dirac cone under magnetic field, notice that the creation and annihilation operators switch on off-diagonals, then we have generally

$$|\Psi_n\rangle = \begin{pmatrix} c_{n1}|n\rangle \\ c_{n2}|n-1\rangle \end{pmatrix}, \tag{77}$$

with order of Landau level labels switched. The eigenequation reads

$$\begin{pmatrix} M\left[\frac{\hbar^2}{l_B^2}(2n+1)\right] - E_n & \eta\sqrt{n} \\ \eta\sqrt{n} & -M\left[\frac{\hbar^2}{l_B^2}(2n-1)\right] - E_n \end{pmatrix} \begin{pmatrix} c_{n1} \\ c_{n2} \end{pmatrix} = 0, \tag{78}$$

and if we continue to use notations of ω_n and m_n defined before, we obtain

$$\varepsilon_{ns} = -\frac{\omega_n}{2} + s\sqrt{m_n^2 + \eta^2 n} = \begin{cases} s\eta\sqrt{n}, & n < n_c \\ -\frac{\omega}{2} + s\sqrt{(n-n_c)^2\omega^2 + \eta^2 n}, & n > n_c \end{cases}, \tag{79}$$

together with the solved wavefunction

$$\varphi_{nsk_x}(r) = \frac{e^{ik_x x}}{\sqrt{L_x l_B}} \begin{pmatrix} \cos\frac{\theta_{ns}}{2}\phi_n(\frac{y+y_0}{l_B}) \\ s\sin\frac{\theta_{ns}}{2}\phi_{n-1}(\frac{y+y_0}{l_B}) \end{pmatrix}. \tag{80}$$

One needs also to notice the $n = 0$ case.

Following the same procedure, with careful notice about the signs involved, we obtain similarly the clean limit quantity

$$\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns} - \varepsilon_{n+1,s'}}\right]^2 = \frac{1}{4}\left[(2n+1)\left(1 + ss'\frac{m_n m_{n+1}}{E_n E_{n+1}}\right) + s\frac{m_n}{E_n} + s'\frac{m_{n+1}}{E_{n+1}}\right] + \frac{ss'}{2}\frac{\eta^2 n(n+1)}{E_n E_{n+1}},$$

which leads to

$$\sum_{s'}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns} - \varepsilon_{n+1,s'}}\right]^2 = \frac{1}{2}\left[2n+1+s\frac{m_n}{E_n}\right], \tag{81a}$$

$$\sum_{s}\left[\frac{\sqrt{2}\hbar}{l_B}\frac{v_{ns,n+1s'}}{\varepsilon_{ns} - \varepsilon_{n+1,s'}}\right]^2 = \frac{1}{2}\left[2n+1+s'\frac{m_{n+1}}{E_{n+1}}\right], \tag{81b}$$

where the sign before s, s' has been reversed compared to $B > 0$ case. Finally, the Hall conductance reads

$$\begin{aligned}\sigma_{xy}(B) &= \mathrm{Im}\,\frac{e^2\hbar}{S}\sum_{\lambda,\lambda'}[n_F(\varepsilon_\lambda-\mu)-n_F(\varepsilon_{\lambda'}-\mu)]\frac{\langle v_x\rangle_{\lambda\lambda'}\langle v_y\rangle_{\lambda'\lambda}}{(\varepsilon_\lambda-\varepsilon_{\lambda'})^2}\\ &= -\frac{e^2}{h}\sum_{s=-}n_F(\varepsilon_{0s}-\mu)+\frac{e^2}{h}\frac{1}{2}\sum_s n_F(\varepsilon_{1,s}-\mu)\left(1+s\frac{m_1}{E_1}\right)\\ &\quad -\frac{e^2}{h}\frac{1}{2}\lim_{N\to+\infty}\sum_{n=1}^{N}\sum_s\left[n_F(\varepsilon_{ns}-\mu)\left(2n+1+s\frac{m_n}{E_n}\right)-n_F(\varepsilon_{n+1,s}-\mu)\left(2n+1+s\frac{m_{n+1}}{E_{n+1}}\right)\right].\end{aligned} \tag{82}$$

The Hall conductance with non-vanishing disorder strength can also be calculated by similar substitution.

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